

Extending Ontology-Driven Validation in Digital Twin Systems for Safe and Trustworthy GeoAI

A Semantically Constrained Decision Framework for Public Asset Management

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ABSTRACT

Digital Twin systems for public asset management increasingly rely on structured and reliable data; however, the transition from validated data to consistent decision-making remains a critical challenge. While ontology-driven validation frameworks ensure data integrity at the point of acquisition, they do not inherently guarantee safe and coherent decision processes, particularly in the context of emerging GeoAI applications.

This paper extends ontology-driven validation approaches by introducing a semantically constrained decision framework within Digital Twin environments. Building upon a previously established deterministic validation structure, the proposed approach incorporates rule-based decision logic and controlled data integration mechanisms to ensure that diagnostic outputs remain consistent, traceable, and aligned with domain-specific knowledge.

Rather than enabling fully autonomous AI-driven decision-making, the framework positions GeoAI as an assistive and bounded layer operating within predefined semantic constraints. This architecture prevents invalid inference propagation, mitigates decision inconsistencies, and reinforces the link between structured inspection data and decision-support processes.

The proposed framework is implemented in a BIM 6D-based Digital Twin for public asset monitoring, demonstrating its applicability in real-world infrastructure inspection and decision-support scenarios. The results indicate that integrating semantic validation with constrained decision logic enhances data reliability, improves decision consistency, and provides a controlled environment for the deployment of GeoAI.

This work contributes to the advancement of trustworthy GeoAI-enabled Digital Twin systems by ensuring that AI operates within semantically bounded conditions. By bridging the gap between data validation and decision-making, the proposed approach enables transparent, auditable, and accountable AI-assisted processes, establishing a robust foundation for safe and reliable public asset management.

Keywords: Digital Twin, GeoAI, Ontology-Driven Validation, Decision-Support Systems, Trustworthy AI, Public Asset Management

1. INTRODUCTION

The increasing adoption of Digital Twin systems in the built environment has enabled new paradigms for infrastructure monitoring, asset management, and data-driven decision-making. By integrating physical assets with digital representations, Digital Twins provide a continuous and structured flow of information throughout the lifecycle of built assets, supporting more efficient and informed management processes (Tao et al., 2019; Deng et al., 2021). However, the reliability of these systems is fundamentally dependent on the quality, consistency, and structure of the underlying data. In the context of building inspection and pathology diagnosis, traditional practices are often characterized by unstructured data collection and subjective interpretation, leading to inconsistencies and limited interoperability across digital systems (Zhao et al., 2022). These limitations compromise the use of inspection data as a reliable basis for decision-making within Digital Twin environments.

Ontology-driven validation frameworks have been proposed as a solution to ensure semantic consistency and data integrity at the point of acquisition. By constraining the relationships between building elements, observable symptoms, and associated pathologies, such approaches transform inspection processes into structured and knowledge-driven systems. Previous research on ontology-driven validation frameworks for building pathology diagnosis, submitted to the CIB W78 conference and currently under review, has demonstrated the effectiveness of deterministic validation mechanisms in improving data reliability. Despite these advances, a critical gap remains between validated data and reliable decision-making. While ontology-based approaches ensure that collected data is semantically consistent, they do not inherently define how this data should be used within decision-support processes. This limitation becomes particularly relevant with the emergence of Geospatial Artificial Intelligence (GeoAI), which introduces new capabilities for pattern recognition, inference, and data-driven analysis within spatially referenced environments, but also raises concerns related to transparency, reliability, and decision accountability.

In this context, the integration of GeoAI into Digital Twin systems requires a structured and controlled framework to prevent the propagation of invalid or inconsistent inferences. Unconstrained AI-based decision models may amplify existing inconsistencies or introduce new forms of uncertainty, particularly when operating on complex and heterogeneous datasets. To address this challenge, this paper extends ontology-driven validation by introducing a semantically constrained decision framework within Digital Twin systems. The proposed approach is based on a dual-layer architecture that integrates spatial context definition, ontology-based validation, and rule-based decision logic. In particular, the framework evolves from a sequential validation model to a constraint-based mechanism in which diagnostic outcomes are derived from the combined relationship between Element and Symptom, ensuring that all possible decisions remain technically valid.

The framework is implemented within a BIM 6D-based Digital Twin environment for public asset management, where each building is georeferenced and inspection data is structured through a dependency-based spatial model. This integration ensures that all inspection records are both spatially contextualized and semantically validated, establishing a coherent foundation for decision-support processes. The main contribution of this work lies in bridging the gap between data validation and decision-making in Digital Twin systems by providing a controlled environment for GeoAI integration. By constraining AI-assisted processes through semantic rules and validated relationships, the proposed framework enables transparent, auditable, and reliable decision support, contributing to the development of safe and trustworthy GeoAI-enabled Digital Twin systems for public asset management.

2. PROBLEM STATEMENT

Despite recent advances in Digital Twin systems for infrastructure monitoring, a critical limitation persists in the transition from structured data acquisition to reliable decision-making processes. While ontology-driven validation frameworks have significantly improved data consistency and semantic coherence in building inspection workflows, they do not inherently provide mechanisms to ensure that decision outcomes derived from such data are logically consistent and technically valid.

Previous approaches have primarily focused on controlling data input through deterministic validation chains, ensuring that inspection records adhere to predefined semantic relationships. However, these models remain fundamentally sequential and limited in their ability to manage multi-condition dependencies, particularly when diagnostic outcomes depend on combined contextual factors rather than single-step relationships. In addition, conventional spatial data structures within inspection systems often rely on static hierarchical representations that lack dependency-based constraints and do not fully exploit the integration between spatial context and semantic validation. This limitation is particularly evident in systems where spatial identification is decoupled from decision logic, leading to potential inconsistencies between location context and diagnostic outputs.

The increasing incorporation of Geospatial Artificial Intelligence (GeoAI) into Digital Twin environments further amplifies these challenges. While GeoAI introduces advanced capabilities for data analysis and inference, its application within unstructured or weakly constrained decision frameworks may result in the propagation of inconsistent or semantically invalid conclusions. This raises critical concerns regarding reliability, transparency, and decision accountability in AI-assisted systems. The absence of integrated mechanisms that simultaneously constrain spatial context, semantic relationships, and decision logic limits the ability to develop robust, scalable, and trustworthy decision-support systems. Without such integration, even high-quality validated data may lead to inconsistent or unreliable outcomes when used for decision-making.

Therefore, there is a clear need for a unified framework that extends ontology-driven validation beyond data acquisition and incorporates constraint-based decision mechanisms within a spatially consistent Digital Twin environment. Such a framework must ensure that diagnostic outcomes are not only semantically valid but also derived from controlled and context-dependent relationships, enabling the safe and accountable integration of GeoAI in public asset management systems.

3. METHODOLOGY

This section presents the proposed semantically constrained framework for integrating spatial context, ontology-driven validation, and decision-support mechanisms within Digital Twin systems. The methodology is structured around a dual-layer architecture that combines a dependency-based spatial model with an ontology-driven validation process, forming the foundation for controlled and reliable decision-making.

3.1 Spatial Context Definition

The inspection process begins with the structured identification of the spatial context within the building asset. Each building is georeferenced through the assignment of geographic coordinates (latitude and longitude), ensuring that all inspection data is inherently linked to a real-world spatial framework.

The spatial hierarchy is defined through a dependency-based sequence:

**Building → Block → Floor → (Context-Dependent Floor Plan
Visualization) → Macrozone → Microzone**

Unlike conventional static hierarchies, the proposed framework enforces strict dependency relationships at each level. The selection of a Block is conditioned by the selected Building, Floors are constrained by the corresponding Building and Block, and Macrozones are defined according to the spatial configuration of each Floor. Microzone represents the most granular spatial unit and is fully dependent on the hierarchical chain of Building, Block, Floor, and Macrozone. Each Microzone is uniquely identified through a coded spatial reference, enabling precise localization, traceability, and reproducibility of inspection records. Upon selection of the Floor, the system dynamically generates a visual representation of the building layout through a floor plan interface. This integrated visualization allows users to interact directly with the spatial model, ensuring accurate identification of Macrozone and Microzone within the asset.

This multi-level dependency structure establishes a deterministic spatial validation chain, where each selection constrains the set of valid subsequent choices. As a result, invalid spatial combinations are inherently prevented, ensuring contextual consistency and reinforcing the integration between spatial, semantic, and operational layers within the Digital Twin environment, consistent with structured Digital Twin architectures that rely on controlled data dependencies (D'Amico et al., 2020).

Similarly, each Microzone is associated with a unique spatial identifier displayed within the floor plan interface. This coded representation provides an additional validation layer, supporting the user in accurately selecting the intended inspection location. Although the system enforces dependency-based constraints that restrict invalid selections, the visual identification of Microzone codes enhances operational reliability by reducing the likelihood of user selection errors. This combination of automated validation and user-assisted confirmation introduces a dual-layer validation mechanism, integrating system-driven constraints with human-in-the-loop verification.

3.2 Ontology-Driven Validation

Following the definition of the spatial context, the diagnostic process is conducted through an ontology-driven validation mechanism that ensures semantic consistency and technical coherence in all inspection records.

The proposed framework builds upon previous ontology-based validation approaches by introducing a constraint-based validation model that extends beyond sequential dependencies. While traditional models follow a linear validation chain, the present approach incorporates a combined dependency structure in which diagnostic outcomes are derived from multiple correlated inputs.

Specifically, the validation process is defined as:

(Element + Symptom) → Pathology

In this formulation, Pathologies are not inferred from isolated inputs but are constrained by the combined relationship between a selected building Element and its associated Symptom. This dual-condition dependency ensures that all diagnostic outputs are restricted to predefined and technically validated combinations, preventing semantically inconsistent or incompatible associations.

The validation mechanism is implemented through a controlled ontology that encodes domain-specific knowledge, defining explicit relationships between Elements, Symptoms, and Pathologies. When the user selects an Element, the system dynamically filters the relevant Symptoms. Subsequently, the selection of a Symptom further constrains the set of valid Pathologies based on predefined semantic correlations. This approach represents a significant evolution from deterministic sequential validation models by introducing a condition-based inference structure. Rather than allowing independent or loosely constrained selections, the system enforces a closed semantic framework in which all possible diagnostic combinations are prevalidated. The integration of ontology-

driven validation with the dependency-based spatial model ensures that each inspection record is simultaneously spatially contextualized and semantically validated. This guarantees that diagnostic outputs are both geographically consistent and technically coherent.

Furthermore, the constraint-based validation mechanism eliminates subjective interpretation at the point of data entry, reducing variability across inspections and ensuring consistency in data representation. This aligns with structured data integration approaches in Digital Twin systems, where controlled semantic relationships are essential for achieving interoperability and reliable decision-support processes (Boje et al., 2020). By constraining both spatial and semantic dimensions, the framework establishes a validated data environment that supports reproducibility, traceability, and consistency. This provides the necessary foundation for higher-level decision mechanisms and enables the safe integration of GeoAI within the Digital Twin context.

3.3 Decision Framework and GeoAI Integration

Building upon the spatial dependency model and ontology-driven validation, the proposed framework introduces a semantically constrained decision layer that enables controlled and consistent decision-support within Digital Twin systems. While ontology-driven validation ensures that all inspection data is semantically valid, it does not inherently define how this data is transformed into reliable decision outcomes. To address this limitation, the proposed approach extends validation mechanisms into a decision framework in which all diagnostic outputs are derived from constrained and context-dependent relationships.

The decision framework is based on a rule-based structure in which validated diagnostic combinations are processed within predefined semantic boundaries. By enforcing that all decisions originate from previously validated (Element + Symptom) relationships, the system ensures that decision outcomes are technically coherent and reproducible. Unlike conventional AI-driven systems that rely on data-driven inference with minimal constraints, the proposed approach introduces a bounded decision environment in which inference is strictly controlled by semantic rules and validated relationships. This prevents the propagation of inconsistent or invalid conclusions and ensures that all outputs remain aligned with domain-specific knowledge. This constraint-based approach enables the integration of GeoAI as an assistive layer rather than an autonomous decision-making system, operating on structured, validated, and spatially contextualized data. Within this framework, GeoAI enables enhanced pattern recognition and analytical capabilities while preserving control, transparency, and reliability.

The integration of spatial context, ontology-driven validation, and constrained decision logic establishes a unified framework in which data, context, and inference are tightly coupled, enabling a controlled and consistent environment for AI-assisted geospatial analysis within Digital Twin systems. The proposed framework introduces a governance-oriented approach to GeoAI deployment by ensuring that all decision-support processes are transparent, auditable, and reproducible. By constraining both input data and inference mechanisms, the system mitigates key risks associated with AI-based systems, including inconsistency, lack of interpretability, and uncontrolled decision propagation.

The proposed methodology is implemented within the PPMI Digital Twin, a BIM 6D-based platform for public asset monitoring, providing a real-world environment for the integration and validation of spatial, semantic, and decision-support processes. By combining dependency-based spatial context definition, ontology-driven validation, and constrained decision logic, the framework establishes a unified and controlled environment in which data, context, and inference are tightly coupled. As a result, the proposed approach enables the safe integration of geospatial AI capabilities within Digital Twin systems, supporting reliable, transparent, and auditable decision-support processes while maintaining full semantic and spatial integrity in public asset management applications.

The overall integration of spatial context, ontology-driven validation, and constrained decision logic is illustrated in Figure 1.

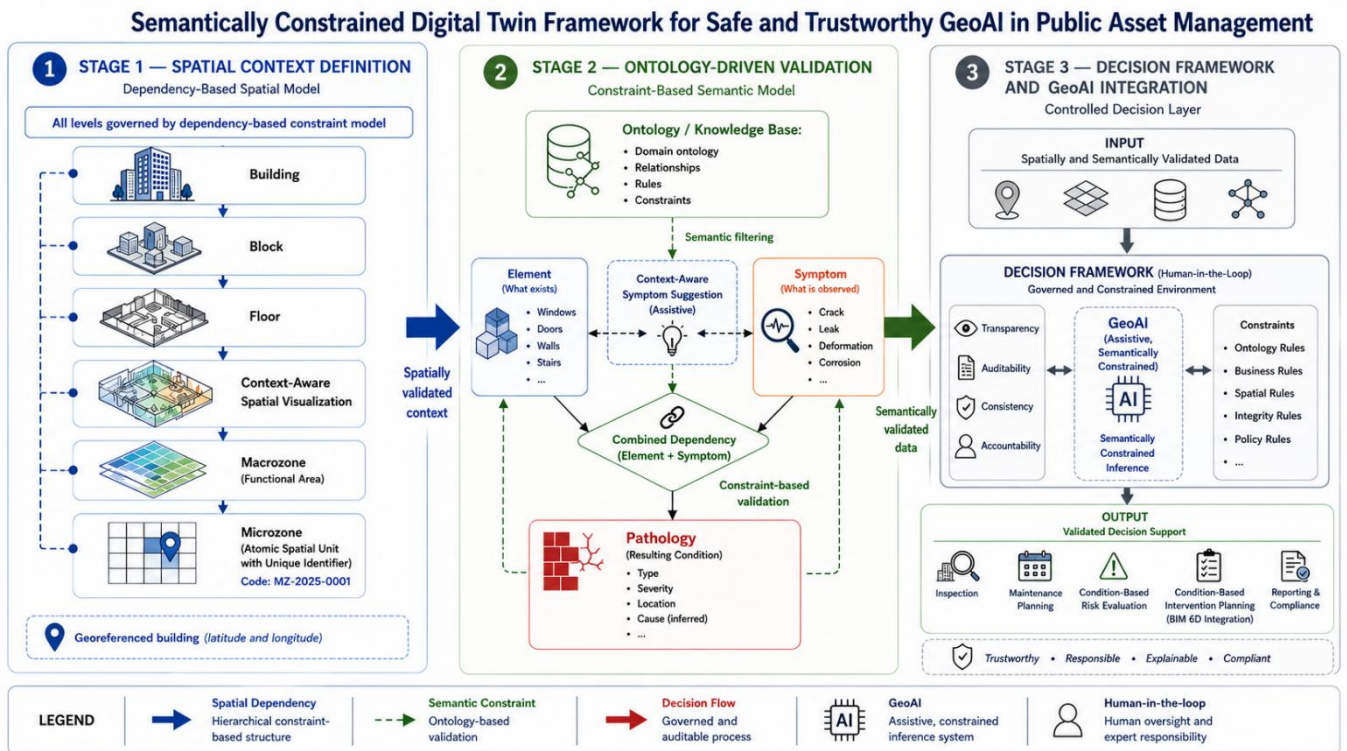


Figure 1: Semantically constrained Digital Twin framework integrating spatial dependency modelling, ontology-driven validation ((Element + Symptom) → Pathology), and a controlled decision framework supporting safe and trustworthy GeoAI in public asset management.

4. RESULTS

The implementation of the proposed framework within the PPMI Digital Twin results in a fully structured inspection workflow in which spatial context, semantic validation, and decision processes are tightly controlled. The dependency-based spatial model ensures that all inspection records are contextually valid, eliminating incorrect spatial associations. The integration of ontology-driven validation further guarantees that only technically consistent Element–Symptom–Pathology relationships are allowed.

4.1 Improved Data Consistency and Validation

The introduction of the combined dependency model ((Element + Symptom) → Pathology) significantly enhances the consistency of diagnostic records. By constraining possible associations between inspection variables, the system eliminates semantically invalid combinations at the point of data entry. This ensures that all recorded diagnoses are technically coherent and aligned with predefined domain knowledge.

Accordingly, inspection data is no longer dependent on individual interpretation but is generated through a controlled validation process. This leads to a standardized data structure that supports consistent reporting and reliable aggregation across multiple assets and inspection scenarios.

4.2 Reduction of User-Induced Errors

The proposed framework introduces a dual-layer validation mechanism that combines system-imposed constraints with user-assisted guidance. The dependency-based spatial structure, together with ontology-driven filtering, restricts user input to valid options at each stage of the inspection process. Additionally, the system incorporates a guided interaction mechanism that assists users throughout the inspection process by structuring the sequence of actions and reducing search complexity. This built-in guidance ensures that users follow a consistent and validated workflow, minimizing the likelihood of incorrect inputs.

The integration of context-aware symptom suggestion further supports the identification of relevant observable conditions, enhancing both efficiency and reliability. This guided interaction minimizes variability between inspectors and ensures that all diagnostic processes follow a consistent and reproducible workflow.

This approach incorporates an embedded AI-assisted support agent accessible directly within the inspection application interface. This functionality allows users to interact with a knowledge-driven assistant to clarify diagnostic steps, interpret system outputs, and resolve uncertainties during the inspection process.

4.3 Spatial and Semantic Integration

The coupling of georeferenced spatial context with ontology-based validation establishes a unified framework in which inspection data is both geographically precise and semantically structured. The dependency-based spatial hierarchy ensures that each inspection record is associated with a valid spatial context, while the ontology enforces consistency across diagnostic relationships.

The use of Microzone as atomic spatial units further enhances this integration by enabling precise localization of conditions within the asset. This allows inspection data to be accurately mapped within the Digital Twin environment, improving traceability and enabling consistent spatial analysis.

4.4 Decision-Support Capabilities

Enhancing validation mechanisms within a rule-based decision framework, the system enables consistent and controlled decision-support processes. All outputs are derived from validated spatial and semantic relationships, ensuring that decisions remain aligned with technical knowledge and predefined constraints. The framework supports condition-based evaluation and intervention planning by linking diagnostic outcomes with structured decision logic. This allows the system to provide justified and consistent recommendations while maintaining full control over the inference process. As a result, decision-support processes are no longer based on subjective interpretation but are grounded in a structured, reproducible, and semantically constrained environment.

4.5 Standardized and GeoAI-Ready Inspection Workflows

The framework achieves standardization of inspection workflows by enforcing a structured and constraint-based data acquisition process. By guiding users through predefined spatial and semantic relationships, it reduces variability in data collection and minimizes dependence on individual interpretation.

Consequently, inspection processes become reproducible and consistently aligned with defined technical criteria. This structured approach ensures that all inspection data is generated within a controlled and validated environment, providing a reliable foundation for AI-assisted analysis. By ensuring consistency across spatial context, semantic validation, and diagnostic relationships, the framework establishes a GeoAI-ready data structure, enabling the integration of geospatial AI

capabilities within a controlled and trustworthy environment, where all inputs are validated and decision pathways remain constrained.

5. DISCUSSION

The results demonstrate a clear shift from conventional inspection systems towards a structured and semantically controlled Digital Twin environment. Traditional inspection approaches are typically characterized by unregulated data collection and a high degree of subjectivity, leading to inconsistent diagnostic records and limited interoperability across datasets. In contrast, the proposed framework enforces a constraint-based workflow in which all inspection data is generated within predefined spatial and semantic boundaries. This structured approach significantly reduces ambiguity and variability, enabling the production of consistent and reliable information across different users and building assets.

A key contribution of this work lies in the integration of dependency-based spatial modelling with ontology-driven validation, forming a unified data structure within the Digital Twin (Aivaliotis et al., 2019). By combining spatial context with semantic constraints, the framework ensures that all inspection records are both geographically valid and technically coherent. In particular, the transition from a sequential validation model to a combined dependency mechanism ((Element + Symptom) → Pathology) represents a significant advancement. This approach reduces the likelihood of incorrect diagnostic associations and introduces a higher level of control over data integrity.

The implications of these findings are particularly relevant for the integration of GeoAI within Digital Twin environments. Conventional AI-based systems often rely on unconstrained data-driven inference, which may produce non-interpretable or inconsistent outputs when applied to heterogeneous inspection datasets. In contrast, the proposed framework establishes a controlled environment in which GeoAI operates on structured, validated, and spatially contextualized data. By constraining inference within predefined semantic relationships, the system ensures that all AI-assisted outputs remain transparent, auditable, and aligned with domain knowledge, supporting the principles of explainable and trustworthy AI within GeoAI systems (Miller, 2019).

In addition, the incorporation of guided interaction mechanisms and AI-assisted user support enhances the inspection process by reducing dependency on individual expertise while maintaining human control. This human-in-the-loop approach reinforces the reliability of the system and ensures that decision-making remains interpretable and accountable, which is particularly critical in public asset management contexts (Zhao et al., 2022).

The framework is currently being implemented within a pilot building managed by the Regional Secretariat for Tourism, Mobility and Infrastructure (SRTMI) within the Regional Government of the Azores. This real-world deployment provides an operational environment for validating the PPMI Digital Twin platform and its BIM 6D-based architecture. The pilot implementation demonstrates the feasibility of integrating spatial, semantic, and decision-support mechanisms under real inspection conditions. Moreover, the structured nature of the framework enables its scalability across multiple buildings. The use of standardized and constraint-based workflows supports the extension of the system to a distributed set of public assets, establishing the foundations for a regional-scale Digital Twin across the Azores archipelago.

Despite these advantages, the current approach is primarily based on rule-based validation and qualitative condition assessment. While this ensures high levels of control and reliability, it limits the integration of fully data-driven decision mechanisms. In this context, future developments should focus on incorporating advanced GeoAI capabilities, including predictive analytics, pattern recognition, and data-driven decision recommendation models, as highlighted in recent GeoAI research (Janowicz et al., 2020).

6. CONCLUSION

This paper presented a semantically constrained Digital Twin framework for building pathology diagnosis, addressing key limitations of conventional inspection practices related to data inconsistency and subjective interpretation. By integrating dependency-based spatial context, ontology-driven validation, and a rule-based decision framework, the proposed approach ensures that all inspection data and diagnostic outputs are generated within a controlled and technically coherent environment. A central contribution of this work lies in the transition from sequential validation models to a combined dependency mechanism ((Element + Symptom) → Pathology), which enhances data reliability and supports consistent decision-support processes. The integration of spatial and semantic constraints establishes a unified data structure that improves traceability, interoperability, and reproducibility across inspection workflows.

The framework creates the necessary conditions for the safe and trustworthy integration of GeoAI within Digital Twin systems. By constraining inference within predefined semantic relationships, the proposed approach ensures that AI-assisted processes remain transparent, auditable, and aligned with domain knowledge. This directly addresses current challenges associated with the application of AI in geospatial systems, particularly in terms of trust, accountability, and data integrity. The implementation within the PPMI Digital Twin platform demonstrates the practical applicability of the framework in a real-world public asset management context. The ongoing deployment within the Autonomous Region of the Azores highlights the feasibility of integrating structured inspection processes within a BIM 6D-based environment and supports the scalability of the approach across a distributed portfolio of public buildings. These findings align with foundational principles of Digital Twin development, where the integration between physical assets and their digital counterparts relies on structured, consistent, and continuously validated data flows (Grieves, 2014). At the same time, the proposed framework contributes to more recent advancements in Digital Twin research by emphasizing the role of semantic structuring and controlled decision-making environments as essential prerequisites for the reliable adoption of GeoAI technologies.

Future work will focus on extending the framework towards advanced GeoAI capabilities, including predictive analytics, pattern recognition, and data-driven decision recommendation models. Additionally, the integration of BIM 6D functionalities, such as cost modelling and lifecycle analysis, will enable the transition from controlled decision-support systems to fully predictive and adaptive Digital Twin environments for strategic public asset management.

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8. DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the author used AI-assisted tools to support language refinement and structuring. All content was reviewed and edited by the author, who takes full responsibility for the final version of the manuscript.

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